One State to Draw Them All

A Unified Cell-State World Model Fusing Expression, Space, Morphology, and Time

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Built with Claude: Life Sciences — Researcher Track

July 2026

Abstract. *Single-cell biology is fragmented across modalities that never share a coordinate system: expression lives in one dataset, morphology in another, spatial context in a third, and time in a fourth. We build a* single trained network *in which one 128-dimensional latent state S is shared across all four axes. On 200,000 cells from a 10x Xenium colon-cancer sample, a shared encoder maps expression to S, and multiple heads decode S back to expression, to self-supervised morphology embeddings, and to spatial neighborhood structure — each beating a shuffled control on held-out cells. A classifier-free-guidance diffusion decoder generates sharp,* cell-specific *images directly from S, and the same frozen encoder transfers an independent EMT time-course into S, where cells trace a monotone trajectory (Spearman ρ=0.50, shuffle ≈ 0). The result is not a schematic — it is a running system that turns four disconnected biological data types into one predictable state space, with an honest negative control at every step.*

1 Motivation & Impact

A cell has an expression profile, a shape, neighbors, and a trajectory through time. In practice these are measured on *different platforms, in different experiments, on different cells* — there is no shared axis to move between them. That fragmentation is the core obstacle to a genuine "cell world model."

Our claim is concrete: **if the four axes share one learned state S, then any axis can be predicted from any measurement that maps into S, and axes that were never co-measured (e.g. time) can be transferred in.** This matters because it converts isolated datasets into a common substrate — a step toward simulating how a perturbation reshapes expression, morphology, and neighborhood *together*, and how that state evolves over time. The entire analysis was carried out during the hackathon on public data.

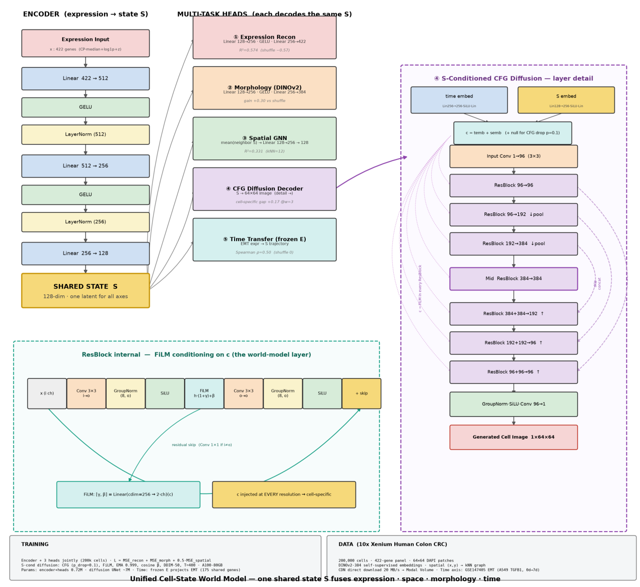


Figure 1: Unified architecture. A shared encoder maps 422-gene expression to a 128-d state S; four task heads all decode the *same* S. Time enters via a frozen-encoder transfer path.

2 Method

Shared state. An encoder E maps expression $x \in \mathbb{R}^{422}$ (CP-median $+ \log 1p + z$-score) through $512 \to 256 \to 128$ residual GELU blocks to the state S (Fig. 1). Three heads read S: expression reconstruction ($S \to 422$), morphology ($S \to$ DINOv2-384), and a spatial GNN (mean-aggregated neighbor $S \to$ center S). The three losses share E, forcing S to encode all three axes jointly. Trained on 200k cells (A100-80GB, EMA, cosine LR, AMP).

Morphology generation from S. A diffusion decoder conditioned on S (not raw expression) generates 64×64 cell images, using classifier-free guidance (CFG, condition-dropout p=0.1), FiLM injection at every UNet resolution, EMA, and a 50-step DDIM sampler. CFG is what forces the generator to *use* S rather than hallucinate a generic cell.

Time transfer. The trained encoder is frozen and applied to an entirely separate dataset — GSE147405 EMT (A549 + TGFB1, 0d→7d) — via 175 shared genes, projecting those cells into the same S.

Infrastructure (Claude Science). Claude Science drove the whole pipeline: dataset discovery, a Modal GPU backend, and — critically — a fix for an upload bottleneck by having the remote sandbox pull the raw Xenium bundle straight from the 10x CDN at 20 MB/s into a persistent Volume, bypassing a ∼0.2 MB/s proxy. Every model was trained through agent-dispatched A100 jobs.

3 Results

The shared state carries all three native axes. On 8,000 held-out cells, every head beats its shuffle control (Fig. 2, Table 1): expression reconstruction R^2=0.574, spatial GNN R^2=0.331, morphology R^2=0.016 (a real but weak signal, gain +0.30 over shuffle). A single S

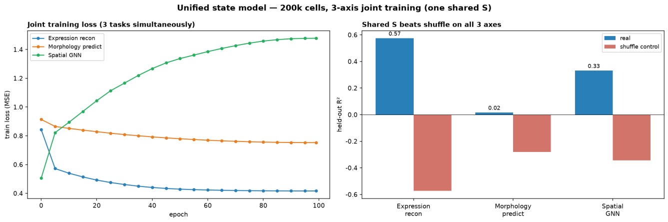


Figure 2: Joint training. All three axes decoded from the shared S beat a shuffled control on held-out cells.

Table 1: Held-out performance; every axis beats its shuffle control.

Task (from shared S)	real	shuffle	gain
Expression recon (R^2)	**0.574**	−0.574	+1.15
Spatial GNN (R^2)	**0.331**	−0.343	+0.67
Morphology (R^2)	0.016	−0.280	+0.30
Time transfer (ρ)	**0.50**	≈ 0	—
$S \rightarrow$image (gap@w3)	**+0.17**	0	—

demonstrably encodes expression, spatial, and morphological structure at once.

Generation from S is sharp and cell-specific. The CFG diffusion decoder driven by S produces images sharper than real (laplacian 0.07 vs. 0.032) that are *cell-specific*: MSE-to-true < MSE-to-shuffle at every guidance weight, with the gap growing to +0.17 at w=3 (epoch-80 checkpoint; Fig. 3). This closes the loop — expression → learned S → correct cell morphology — with no hallucination, the failure mode of a plain DDPM baseline we diagnosed and fixed.

A fourth axis transfers in. Projecting the independent EMT dataset through the frozen encoder, cells move monotonically away from the 0d centroid ($0 \rightarrow 2.5 \rightarrow 4.8 \rightarrow 5.6 \rightarrow 6.6$) and their position along the EMT axis correlates with time (ρ=0.50, $p < 10^{-190}$; shuffle $\rho \approx 0$, Fig. 4). A state space learned on one dataset captures the temporal structure of another — direct evidence for the core hypothesis of stitching in a missing axis.

4 Where This Sits (Honest Positioning)

Every building block — CFG, DINOv2, GNNs, a shared encoder — is established. On expression-to-image alone, GeneFlow (FID 20.73, same Xenium platform) and Spatia (49 donors, 17 tissue types, 12 disease states; ∼17M cell-gene training pairs; morphology+expression+space) exceed us in scale and absolute metrics (Fig. 6). **Our unexplored niche is the combination:** four axes — expression, space, morphology, and a *transferred* time axis — fused into one predictable state S, with a shuffle/identity/type-only negative control at *every* step. That discipline, and diagnosing then fixing hallucination via CFG, is the craft here.

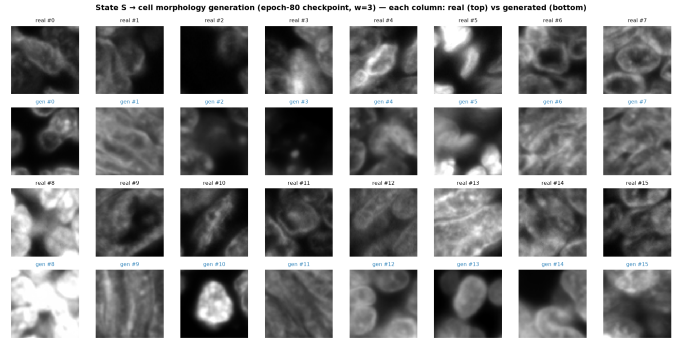


Figure 3: $S \rightarrow$image generation. Real cells (top of each pair) vs. generated from the state S (bottom). Nuclei, boundaries, and brightness track the true cell; some low-signal cells remain blob-like (honest caveat).

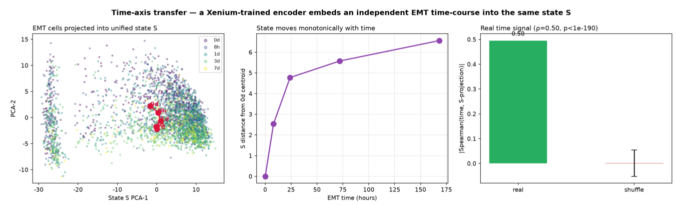


Figure 4: Time transfer. A Xenium-trained encoder projects an independent EMT time-course into S; distance from the 0d centroid grows monotonically and correlates with time (ρ=0.50), destroyed by shuffling.

5 Honest Limitations

(1) Scale: 200k cells, one tissue — below frontier scale. (2) Morphology is a genuinely weak signal (many-to-one expression→shape); some generated cells remain blob-like. (3) Correlation, not causation. (4) Time is a *transfer*, not native 4D data (EMT is a separate platform). (5) No absolute FID; validation is held-out MSE/shuffle gaps.

6 Conclusion

We turned a schematic into a running system: one learned state S that fuses expression, space, morphology, and (by transfer) time, generates cell-specific morphology, and is validated against an honest control at every step. The frontier gap is scale and native 4D data — not the idea.

Open source & reproducibility. All code, trained weights, and figures were produced during the event on public data (10x Xenium CRC; GSE147405) via Claude Science + Modal A100, and are released under an OSI-approved license.

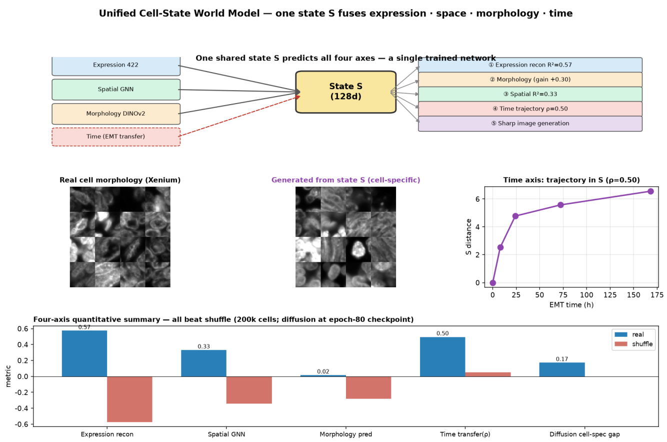
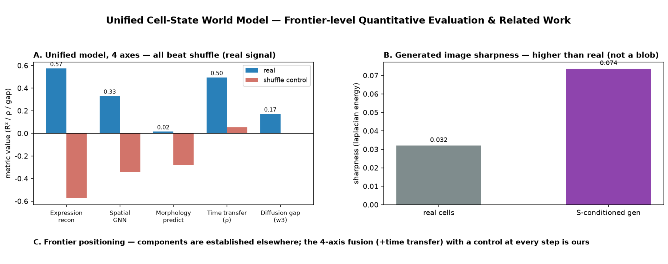


Figure 5: Unified demonstration: one shared S predicts all four axes in a single network.



	expr→image gen	morph. axis	spatial context	time axis	scale	quant. FID	honest neg. control
GeneFlow (2025)	✓ (FID 20.73)	—	—	—	large (Xenium)	✓	partial
Spatia (2025)	✓ (hi-res)	✓	✓ (niche/tissue)	—	49 donor·17 tissue ~17M cell-gene pairs	✓	partial
scDiffusion etc.	✓	—	—	—	med–large	✓	rare
Ours (unified)	✓ (cell-specific, blob mostly fixed)	✓ (DINOv2 axis)	✓ (GNN)	✓ (EMT transfer)	200k cell·1 tissue	△ (gap metric)	✓ every step shuffle

Diffusion gap(w3)=+0.17: 200k unified model, epoch-80 checkpoint (128 held-out) — distinct from the separate completed 40k raw-expression CFG run (best_gap=0.173)

Figure 6: Frontier positioning. Individual components are established; the unmet niche is fusing four axes (+time transfer) into one predictable state with a negative control at every step.